A Major Project Report on

"Eyeball Based Cursor Movement Control Using OpenCV"

Submitted to

OSMANIA UNIVERSITY, Hyderabad

in partial fulfillment of the requirements for the award of degree

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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2023-2024

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CERTIFICATE

This is to certify that the Major Project on "**Eyeball Based Cursor Movement Control Using OpenCV**" is being submitted by the following students:

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They have presented the project work during the academic year 2023–2024 in partial fulfillment of the requirements for the award of **BACHELOR OF ENGINEERING** in **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)** from **Osmania University.** This is a bonafide record work carried out by them under our guidance and supervision. The results of investigation enclosed with this report have been verified and found to be satisfactory.

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DECLARATION

We hereby declare that the project work entitled **"Eyeball Based Cursor Movement Control Using OpenCV"** submitted to the Department of **Computer Science and Engineering (Artificial Intelligence and Machine Learning) of NAWAB SHAH ALAM KHAN COLLEGE OF ENGINEERING AND TECHNOLOGY**, affiliated to Osmania University, Hyderabad in the partial fulfillment of the requirement for award of the degree in BACHELOR OF ENGINEERING in CSE (AI & ML) is a bonafide work done by the undersigned.

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LIST OF ABBREVATIONS

S.NO	ABBREVATION	EXPANSION
1.	AI	Artificial Intelligence
2.	OPENCV	Open-Source Computer Vision (library)
3.	HTML	Hyper Text Markup Language

ABSTRACT

This paper introduces an algorithm to carry out the functions of a mouse by providing a hands-free interaction between humans and computers. It provides an alternative to the traditional mouse computer. By using different expressions of a face using computer vision and matching it with already stored expression and execute actions as per the move. This algorithm will help physically disabled people to perform the functions of a mouse using their face and eye movement. It allows them to left-click, right-click, scroll up and down, move the cursor up, down, left, right. The system has a very basic need like webcam, NumPy, OpenCV, and a few other basic libraries. Different methods have been introduced previously and these methods require the use of attachments and electrodes means some physical connection is needed to the head, which makes them impractical. The method described in this paper does not use any of this expensive equipment. Also, even no physical connection has to be made with the user through any equipment. It just needs a computer with a webcam making it easy and cheap and uncomplicated method.

CHAPTER 1
INTRODUCTION

1.1 GENERAL

The advent of computers in 1938 has changed the way we live our lives today at ease. They have gone through rigorous changes. Initially, their task was to solve mathematical problems and word processing now they have travelled a long way to making a necessary space in our lives with its uses ranging from professional to personal such as browsing on the internet, connecting the world through social media, controlling large machines in factories using the right and timely operations. Computers are readily usable to persons who have hands to control mouse and keyboard and eyes to see what is on the monitor. Now even blind can use the computer using the text to speech option which pronounces the content on the screen. But people who do not have hands cannot use it due to their inability to control mouse and keyword operations. In today's competitive environment everybody needs to be equipped with different sets of skills so as get a job. Knowledge and usage to computers being a necessity nowadays. So this causes them to live a miserable life. They have to beg and do pitiful activities to keep their lives going. By introducing them to technology and make them computer compatible will create a ray of hope for them to learn and do some work to their livelihood. Several efforts are made by professionals to assist the disabled in providing them a tool to interact with the computers using signals such as electroencephalography (EEG) from the brain, facial muscle signals (EMG), and electrooculogram (EOG). Also, methods like limbus, pupil, and eye/eyelid tracking, contact lens method, corneal, pupil reflection relationship have also been implemented. These methods require the use of attachments and electrodes means some physical connection is needed to the head, which makes them impractical. The method described in this paper does not use any of this expensive equipment. Also, even physical connection has to be made with the user through any equipment. It just needs a computer with a webcam making it easy and cheap and uncomplicated method. By taking real-time images of the user the program is designed to evaluate these images and identify the type of action to be performed by taking a match against the previously stored expressions to identify the operation to be performed and take parameter if needed in case of moving the cursor the change in the relative position of eyes concerning the central position talks about the direction to move the

cursor. The implementation of this method would open a wide range of options for people who don't have hands. It will enable them to go to school and learn through the computer with the mouse being controlled by their head movement. It would also give them job opportunities in the service to earn them a job.

ABOUT PROJECT DOMAIN

Artificial intelligence is based on the principle that human intelligence can be defined in a way that a machine can easily mimic it and execute tasks, from the simplest to those that are even more complex. The goals of artificial intelligence include mimicking human cognitive activity. Researchers and developers in the field are making surprisingly rapid strides in mimicking activities such as learning, reasoning, and perception, to the extent that these can be concretely defined. Some believe that innovators may soon be able to develop systems that exceed the capacity of humans to learn or reason out any subject. But others remain skeptical because all cognitive activity is laced with value judgment's that are subject to human experience. As technology advances, previous benchmarks that defined artificial intelligence become outdated. For example, machines that calculate basic functions or recognize text through optical character recognition are no longer considered to embody artificial intelligence, since this function is now taken for granted as an inherent computer function. AI is continuously evolving to benefit many different industries. Machines are wired using a cross-disciplinary approach based on mathematics, computer science, linguistics, psychology, and more.

1.2 OBJECTIVE

Basically, if anybody wants to do any operation on computer or laptop, mouse is the necessary hardware required. Whereas controlling mouse without using hand we are developing an application which track the eye and according to the eye moment mouse is controlled. This application allows the user to use the computer system with the help of user's eyesight and move the cursor as per the direction that is given by the user. Aim of this application is to introduce a low cost-based system for the people who have disability and connect them with the real world.

Artificial Intelligence (AI)

Artificial intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions. The term may also be applied to any machine that exhibits traits associated with a human mind such as learning and problem-solving. The ideal characteristic of artificial intelligence is its ability to rationalize and take action that have the best chance of achieving a specific goal. A subset of artificial intelligence is machine learning, which refers to the concept that computer programs can automatically learn from and adapt to new data without being assisted by humans. Deep learning techniques enable this automatic learning through the absorption of huge amounts or unstructured data such as text, images, or video.

Types of Artificial Intelligence

Artificial Intelligence can be divided in various types, there are mainly two types of main categorization which are based on capabilities and based on functionally of AI. Following is flow diagram which explains the types of AI.

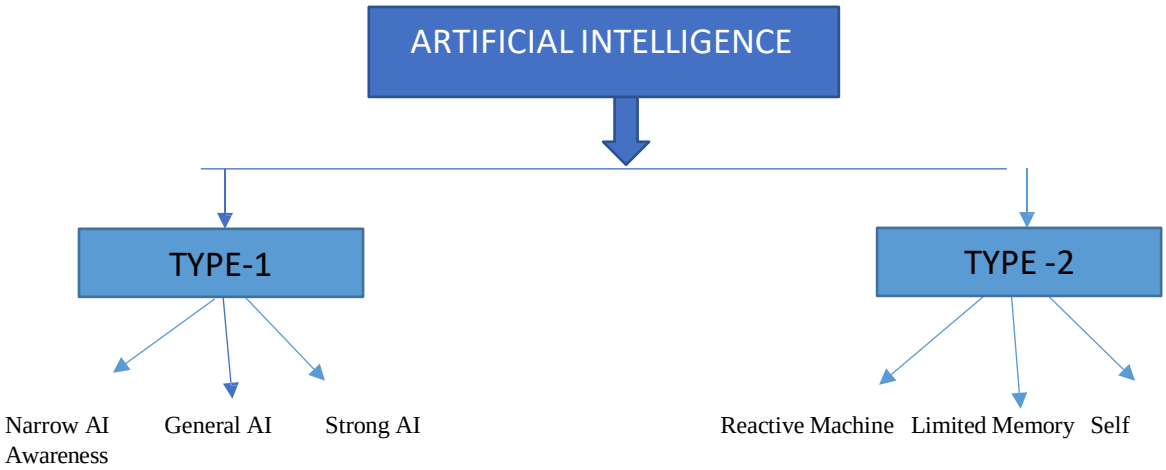


Figure 1.1 Types of Artificial Intelligence

AI type-1: Based on Capabilities

1. Weak AI or Narrow Al

Narrow AI is a type of AI which is able to perform a dedicated task with intelligence. The most common and currently available AI is Narrow AI in the world of Artificial Intelligence.

Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task. Hence it is also termed as weak AI. Narrow AI can fail in unpredictable ways if it goes beyond its limits.

Apple Siris a good example of Narrow Al, but it operates with a limited predefined range of functions.

1BMS watson supercomputer also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing

Some Examples of Narrow AI are playing chess, purchasing suggestions on commerce site, self-driving cars, speech recognition, and image recognition.

2. General AI

General AI is a type of intelligence which could perform any intellectual task with efficiency like a human.

The idea behind the general AI to make such a system which could be smarter and think like a human by its own.

Currently, there is no such system exist which could come under general AI and can perform any task as perfect as a human.

The worldwide researchers are now focused on developing machines with General AI.

As systems with general Al are still under research and it will take lots of efforts and time to develop such systems.

3. Super AI

Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties. It is an

outcome of general Al. Some key characteristics of strong Al include capability include the ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate by its own.

Super Al is still a hypothetical concept of Artificial Intelligence. Development of such systems in real is still world changing task.

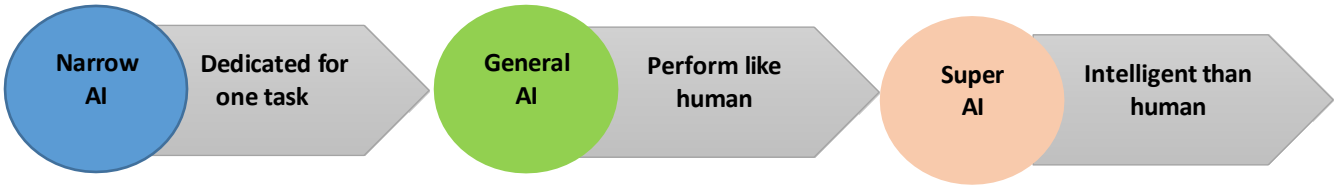


Figure 1.2 Super AI

Artificial Intelligence type-2: Based on functionality

1. Reactive Machines

Purely reactive machines are the most basic types of Artificial Intelligence.

Such Al systems do not store memories or past experiences for future actions.

These machines only focus on current scenarios and react on it as per possible best action.

IBM's Deep Blue system is an example of reactive machines.

Google's Alpha Go is also an example of reactive machines.

2. Limited Memory

Limited memory machines can store past experiences or some data for a short period of time.

These machines can use stored data for a limited time period only.

Self-driving cars are one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

3. Theory of Mind

Theory of Mind AI should understand the human emotions, people, beliefs, and be able to interact socially like humans.

This type or Al machines is still not developed, but researchers are making lots of efforts and improvement for developing such Al machines.

4. Self-Awareness

Self-awareness Al is the future of Artificial Intelligence. These machines Will De super intelligent, and will have their own consciousness, sentiments, and self-awareness.

These machines will be smarter than human mind.

Self-Awareness Al does not exist in reality still and it is a hypothetical concept.

1.3 EXISTING SYSTEM

Now even blind can use the computer using the text to speech option which pronounces the content on the screen. But people who do not have hands cannot use it due to their inability to control mouse and keyword operations. In today's competitive environment everybody needs to be equipped with different sets of skills so as get a job. Knowledge and usage of computers being a necessity nowadays. So this causes them to live a miserable life. They have to beg and do pitiful activities to keep their lives going. By introducing them to technology and make them computer compatible will create a ray of hope for them to learn and do some work to their livelihood.

1.3.1 Existing System Disadvantages

It affects eyes of users and causes problems in vision for them.

It can only handle blinks and is not able to handle short blinks.

Short blinks it simply avoids.

1.4 PROPOSED SYSTEM

The camera keeps looking for pictures at the rate of 30 frames per second. If it can detect a face then it checks whether the face control function is on or off.

Then if it is enabled it captures frames for different instructions by the user through the movement of his face as per the directions given.

If he wants to turn the scroll mode on he/she has to squint both eyes and then move his face up or down for scrolling.

To simply move the cursor user will simply move his head.

For click, he will squint one of his eyes as per the side of the mouse he wants to press.

To disable the mouse control, he has to again open his mouse open.

1.4.1 Proposed System Advantages

Hands-free computing

Facilitating the handicapped using the computer.

Controlling the mouse pointer through eye movement.

Eye based human computer interaction provide real time eye tracking and eye gaze estimation.

Simulate mouse functions

User can control mouse events like left click, right click, and double click etc. by movement of eyes. User can control and trigger mouse using voice commands.

CHAPTER 2
LITERATURE SURVEY

2.1 GENERAL

This chapter gives the Literature Survey about various important techniques and algorithm on using eye tracking and human computer interaction.

2.2 Title: Hand Motion Based Mouse Cursor Control Using Image Processing technique

Author: P. Balamurugan, J. Santhosh.

Year: 2022

Description: The various devices are used for communications between the human and the machine. These devices are keyboard, mouse, etc. for performing tasks. While the workstation technologies are developing up, the roll of Person Workstation Interaction (HC) is also increasing rapidly. Most of the devices uses touch screen technology which cannot be affordable to all the applications. For the traditional touch screen, virtual mouse can be used as substitute way in virtual person interactive module. Motion based interaction systems are very popular at workplace and at home. This paper proposes to develop a method and model that identifies hand motions which can be used as an input command to work together with the workstation system. The motion is recognized and the action specific is performed. While developing the system using image processing, one key area is looking for the system. The method is created to recognize the motions and built in task for every motion. PyAutoGUl is providing coding system in Python and uses he OpenCV library. The purpose of real-time hand motion recognition in the real world are numerous, since it can be used almost anywhere. The technique is focused on the use of a webcam. Then it is used to develop a virtual Person Workstation Interaction Device in a cost-effective manner.

2.3 Title: Human computer interaction-based eye-controlled mouse

Author: Vinay S Vasisht, Swaroop Joshi.

Year: 2021

Description: With advanced technologies in this digital era, there is always scope for development in the field of computing. Hands free computing is in demand as of today it addresses the needs to quadriplegics. This paper presents a Human computer interaction (HC) system that builds of great importance to amputees and those who have issues with using their hands. The system built is an eye based interface that acts as a computer mouse to translate eye movements such as blinking, gazing and squinting towards the mouse cursor actions. The system in discussion makes use of a simple webcam and its software requirements are Python (5.6), OpenCv, numpy and a few other packages which are necessary for face recognition. The face detector can be built using the HOG (Histogram of oriented Gradients) feature along with a linear classifier, and the sliding window technique. It is hands free and no external hardware or sensors are required.

2.4 Title: Real time eye blinking detection and tracking using opencv. Technology

Author: Dhaval Pimplaskar.

Year: 2020

Description: Robust and accurate algorithm in real time eye tracking system has been a fundamental and challenging problem for computer vision. This paper proposed a new method to estimate eye-position and direction based on initial centroid analysis technique. The proposed method was validated by tracking eye position within high and low occlusion condition. In this paper, we represent a methodology for detection of eye blinking robustly in real time environment. Here we use the connected component technique and centroid method to track and blinking of eyes on OpenCV platform which is open source and developed by Intel.

2.5 Title: Real-Time Driver Drowsiness Detection System Using Eye Aspect Ratio and Eye Closure Ratio

Author: Sukrit Mehta

Year: 2019

Description: Every year many people lose their lives due to fatal road accidents around the world and drowsy driving is one of the primary causes of road accidents and death. Fatigue and micro sleep at the driving controls are often the root cause of serious accidents. However, initial signs to fatigue can be detected before a critical situation arises and therefore, detection of driver' s fatigue and its indication is ongoing research topic. Most to the traditional methods to detect drowsiness are based on behavioral aspects while some are intrusive and may distract drivers, while some require expensive sensors. Therefore, in this paper, a light-weight, and real time driver's drowsiness detection system is developed and implemented on Android application. The system records the videos and detects driver's face in every frame by employing image processing techniques. The system is capable of detecting facial landmarks, computes Eye Aspect Ratio (EAR) and Eye Closure Ratio (ECR) to detect driver's drowsiness based on adaptive thresholding. Machine learning algorithms have been employed to test the efficacy of the proposed approach.

2.6 Title: Open face: an open-source facial behavior analysis tool kit

Author: Tadas Baltrušaitis; Peter Robinson

Year: 2016

Description: Over the past few years, there has been an increased interest in automatic facial behavior analysis and understanding. We present Open Face an open-source tool intended for computer vision and machine learning researchers, affective computing community and people interested in building interactive applications based on facial behavior analysis. OpenFace is the first open-source tool capable of facial landmark detection, head pose estimation, facial action unit recognition, and eye-gaze estimation. The computer vision algorithms which represent the core of OpenFace demonstrate state- of-the-art results in all of the above-mentioned tasks. Furthermore, our tool is capable of real-time performance and is able to run from a simple webcam without any specialist.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 GENERAL

These are the requirements for doing the project. Without using these tools and software we can't do the project. So, we have two requirements to do the project. They are

1. Hardware Requirements.

2. Software Requirements.

3.2 HARDWARE REQUIREMENTS

The hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system. They are used by software engineers as the starting point for the system design. It shows what the system does and not how it should be implemented.

PROCESSOR : DUAL CORE 2 DUOs.

RAM : 4GB DD RAM

HARD DISK : 250 GB

3.3 SOFTWARE REQUIREMENTS

The software requirements document is the specification of the system. It should include both a definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks and tracking the teams and tracking the team's progress throughout the development activity.

OPERATNG SYSTEM : WINDOWS 7/8/10

PLATFORM : SPYDER

PROGRAMMING LANGUAGE : PYTHON

FRONT END : PYCHARM

3.4 FUNCTIONAL REQUIREMENTS

A functional requirement defines a function of a software-system or its component. A function is described as a set of inputs, the behavior, it allows them to left-click, right-click, scroll up and down, move the cursor up, down, left, right. The system has a very basic need like webcam, NumPy, OpenCV, and a few other basic libraries.

- If he wants to turn the scroll mode on he/she has to squint both eyes and then move his face up or down for scrolling.
- To simply move the cursor user will simply move his head.
- For click, he will squint one of his eyes as per the side of the mouse he wants to press.
- To disable the mouse control, he has to again open his mouse open.

3.5 NON-FUNCTIONAL REQUIREMENTS

The major non-functional Requirements of the system are as follows

- **Usability**

The system is designed with completely automated process hence there is no or less user intervention.

- **Reliability**

The system is more reliable because of the qualities that are inherited from the chosen platform java. The code built by using java is more reliable.

- **Performance**

This system is developing in the high-level languages and using the advanced front-end and back-end technologies it will give response to the end user on client system with in very less time.

- **Supportability**

The system is designed to be the cross platform supportable. The system is Supported on a wide range of hardware and any software platform, which is having JVM, built into the system.

- **Implementation**

The system is implemented in web environment using struts framework. The apache tomcat is used as the web server and Windows XP professional is used as the platform. Interface the user interface is based on Struts provides HTML Tag.

CHAPTER 4
METHODOLOGY AND MODULES

4.1 GENERAL

The algorithm starts to detect a face of a person using a face detection algorithm. A detecting the face, the position of eyes and mouth is captured for controlling the mouse functions such as left-click, right-click, and cursor movement. This method doesn't require any special hardware and sensors. It is a hand -free system that is useful for handicapped. First, we have to activate the mouse control mode by opening the mouth. For moving the mouse cursor left or right we simply have to move our head in the intended direction or movement of a cursor. For enabling the scroll mode, we have to squint our eyes into the camera and on initiation; it will show scroll mode is on. Now you can move your head up for scrolling the up and similarly down to scroll the page down.

4.2 PROBLEM DEFINITION

The problem here is people who do not have hands cannot use it due to their inability to control mouse and keyword operations. For disable people to control the mouse here we are developing eye tracking system by that they can easily use the mouse by their face.

4.3 SYSTEM ARCHITECTURE

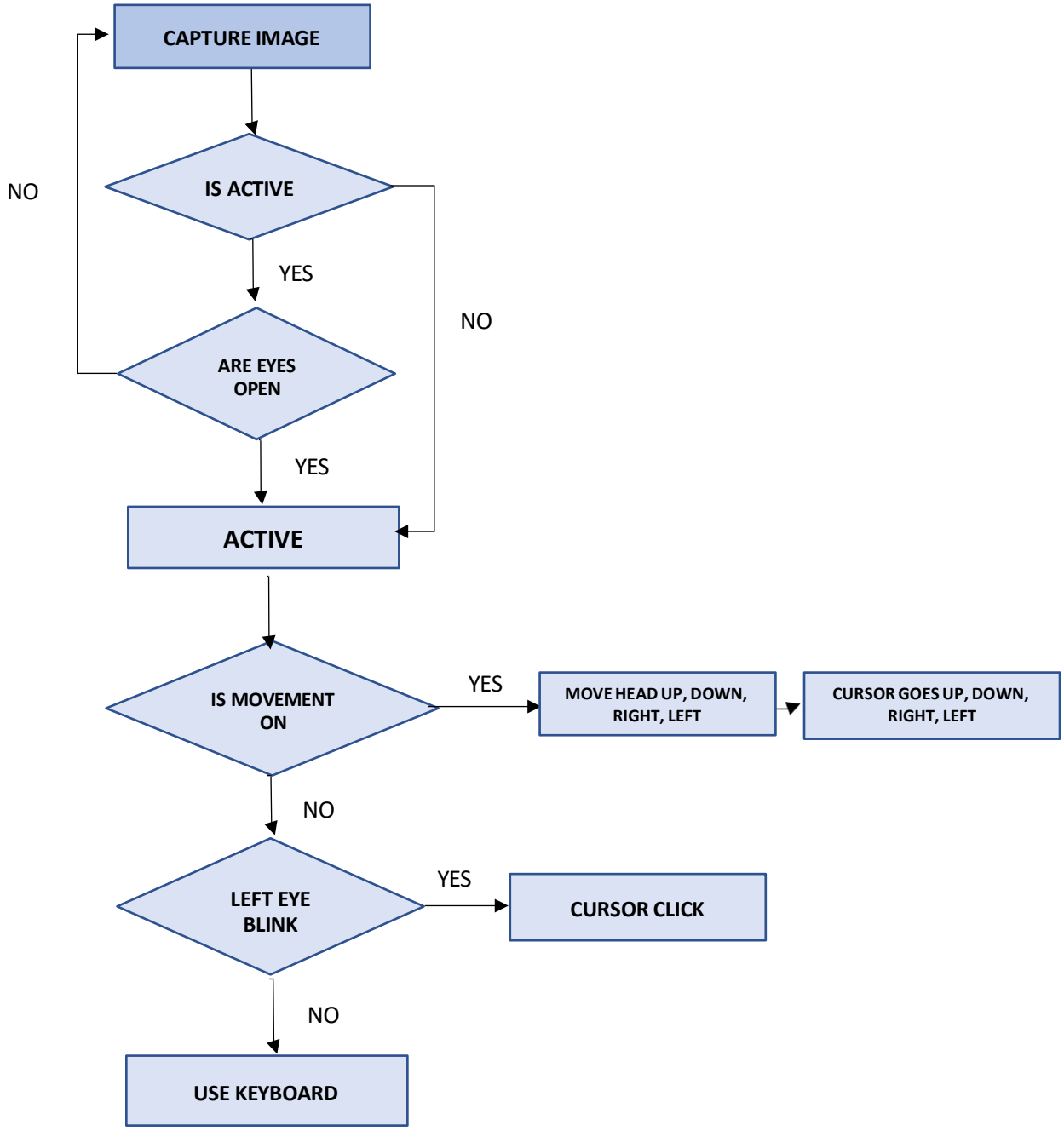


Figure 4.1 System Architecture

4.3 METHODOLOGIES

The method or mouse movement by pupil and mouth movement was achieved using following process:

4.3.1 List of Modules

- Face Detection
- Eye Detection
- Algorithm
- Usage

- **FACE DETECTION**

To represent a clean and accurate image, the user should face parallel to the webcam position as represented. The image of the person is captured by webcam preinstalled in the system using OpenCV and processed using python. It shows the captured image using a webcam. Dlib has a pre-trained dataset iBUG 300-W which has 68 coordinates that mapped to the face of a person. The facial detector detects key landmarks on the face of a person and tracks them. It shows 68 facial coordinates.

- **EYE DETECTION**

For eye detection, we use Eye-Aspect-Ratio (EAR). It was used to observe if the person's eye is flickering or not in the video frame. Each eye is expressed as 6 coordinated (pl-p6), pl is the coordinate of the left part of the eye and then p2-p6 is located accordingly when we travel in the clockwise direction as pl-p6 are points located on the eye p2-pó and p3-p5 depicts the vertical distance eye-aspect-ratio (EAR) is consistent until the time eye is open, but it tends to zero when a flickering is started between the eyes pl-p4 depicts the horizontal distance between the eyes.

- **ALGORITHM**

The algorithm used can be depicted by the flowchart shown in Fig 4.1.

1. The camera keeps looking for pictures at the rate of 30 frames per second. If it can detect face, then it checks whether the face control function is on or off

2. Then if it is enabled it captures frames for different instructions by the User through the movement of his face as per the directions given.

3. 1fhe wants to turn the scroll mode on, he/she has to squint both eyes and then move his face

4. Up or down for scrolling.

5. To simply move the cursor user will simply move his head.

6. For click, he will squint one of his eyes as per the side of the mouse he wants to press.

7. To disable the mouse control, he has to again open0 his mouse open.

8. For most laptop touch pad is not the most comfortable and convenient

9. Removes the requirement of having physical involvement.

10. User friendly application.

After the implementation of the algorithm, it was tested on a webpage and it showed correct results. The test started with the camera being turned on looking to detect a face. As soon it detects a face it marks the border of eyes and mouth to tell the user about the areas it will consider for an input. From the initial position and will draw a line in the direction of movement which will tell about the length traveled from the epicenter. For moving the mouse cursor left or right we simply have to move our head in the intended direction of movement of the cursor. For turning the scroll mode on we have to Squint our eyes into the camera and on initiation, it will show scroll mode is ON Now you can move your head up for scrolling the up and similarly down the page down. For clicks, we have to blink the left eye and right eye for left-click a right-click respectively. They are shown for results. Depicts cursor movement the left direction when mouse control is activated by opening mouth for a certain Depicts cursor movement in the right direction when mouse control 15 activated by opening mouth for a certain time. Depicts scrolling function towards a downward direction when mouse control is activated by opening mouth for a certain time and scrolling mode is also activated by squinting eyes. Depicts scrolling function towards upward direction when mouse control is activated by opening mouth for a certain tum and scrolling mode is also activated by squinting eyes.

- **Convolutional Neural Network**

A convolutional neural network can have tens or hundreds of layers that each learn to detect different features of an image. Filters are applied to each training image at different resolutions, and the output of each convolved image is used as the input to the next layer. The filters can start as very simple features, such as brightness and edges, and increase in complexity to features that uniquely define the object.

Neural network can be divided into two kinds, biological neural network is one of them, and artificial neural network is another kind. Here mainly introduces artificial neural network. An artificial neural network is a data model that processes information and is similar in structure to the synaptic connections in the brain. Neural network is composed of many neurons; the output of the previous neuron can be used as the input of the latter neuron. The corresponding formula is as follows:

This unit is also called Logistic regression model. When many neurons linked together, and when they were layered, the structure can now be called a neural network model. Figure shows a neural network with hidden layers.

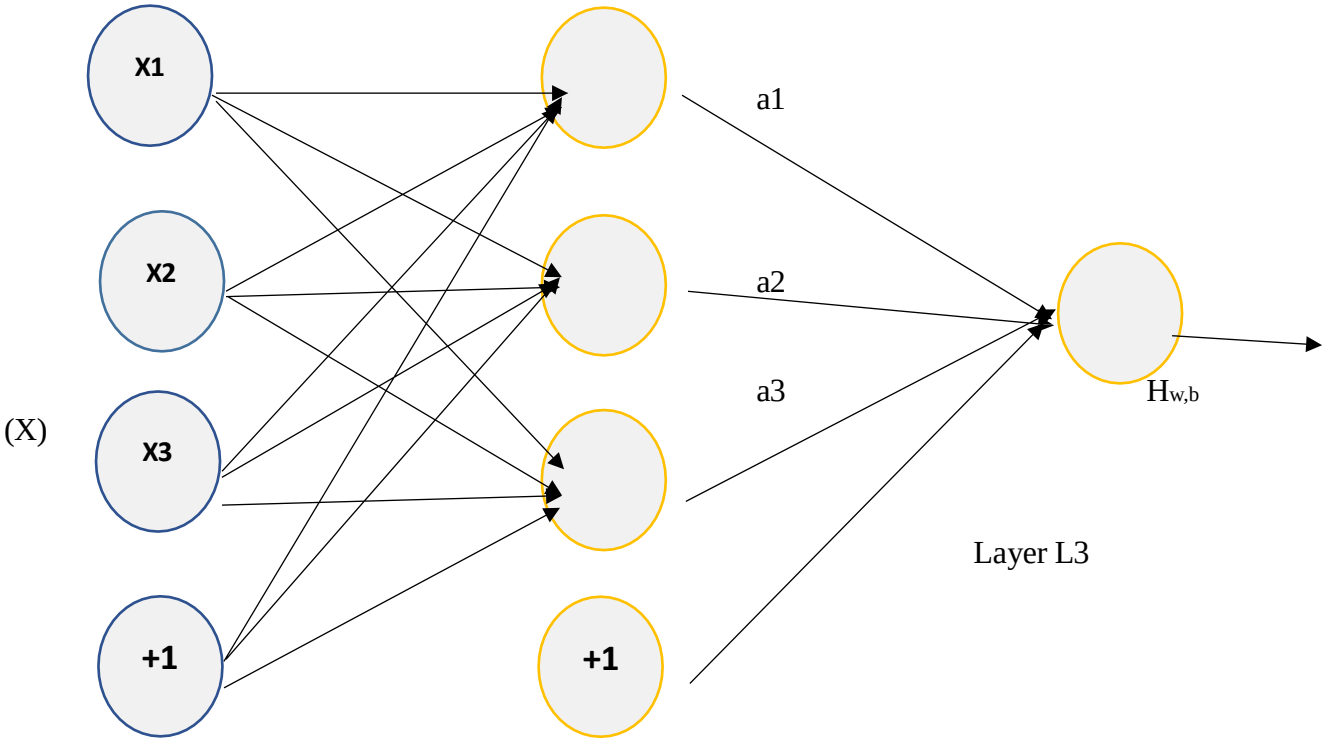


Figure 4.2 Neural Network

- **USAGE**

The explained the different modes and working of the algorithm. In this, we explained the action and their corresponding function we used in our algorithm. For activating/deactivate mouse control we have to open mouth continuously for some times. If the user blinks their left eye it will act as a left click. If the user blinks their right eye it will act as a right-click. If the user squints his/her eye for some time it will activate/deactivate scrolling. Head will be used for cursor movements and scrolling up or down.

CHAPTER 5

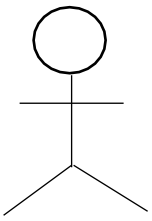
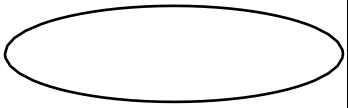
DESIGN ENGINEERING

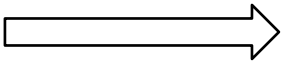
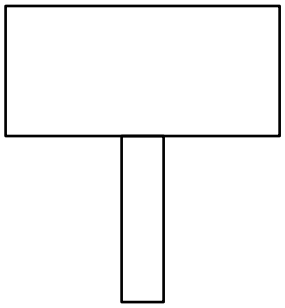
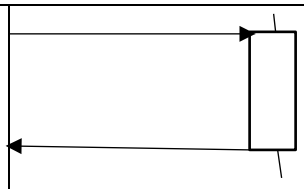
5.1 GENERAL

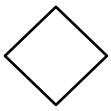
Design Engineering deals with the various UML [Unified Modeling Language] diagrams for the implementation of project. Design is a meaningful engineering representation of a thing that is to be built. Software design is a process through which the requirements are translated into representation of the software. Design is the place where quality is rendered in software engineering. Design is the means to accurately translate customer requirements into finished product.

5.2 UML NOTATION

UML notations are the most important elements in modeling. Efficient and appropriate use of notations is very important for making a complete and meaningful model. Different notations are available for things and relationships. And the UML diagrams are made using the notations of things and relationships.

NAME	SYMBOL	DESCRIPTION
Actor		Internal or external entity that interacts with the system.
Use Case		Use case is used to capture high level functionalities of a system.
Dependency		A semantic relationship between two elements in which a change to one element may affect the meaning of other.

Association		A structural relationship describing a set of links connected between objects.
Generalization		A relationship in which objects of a specialized element (child) are substitutes for objects of a generalized element (parent)
Classifier Role		A Classifier describes a set of instances that have common behavioral and structural features.
Call Action		This action send message and wait for a reply. The reply message is sent back to that message.
Send Action		Create message is sent to a lifeline to create itself.
Return Action		Send message and proceed immediately without waiting for return value.
Classifier Role		A classifier describes a set of instances that have common behavioral and structural features.
Initial State		It shows the starting point of a process.
Choice		Collection of operation that specifies a service of class.

Junction		A diamond represents a decision with alternate paths.
Action		Action states represent the non-interruptible actions of objects.

5.3 UML DIAGRAMS

5.3.1 Use Case Diagram

The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted. The above diagram consists of user as actor. Each will play a certain role to achieve the concept.

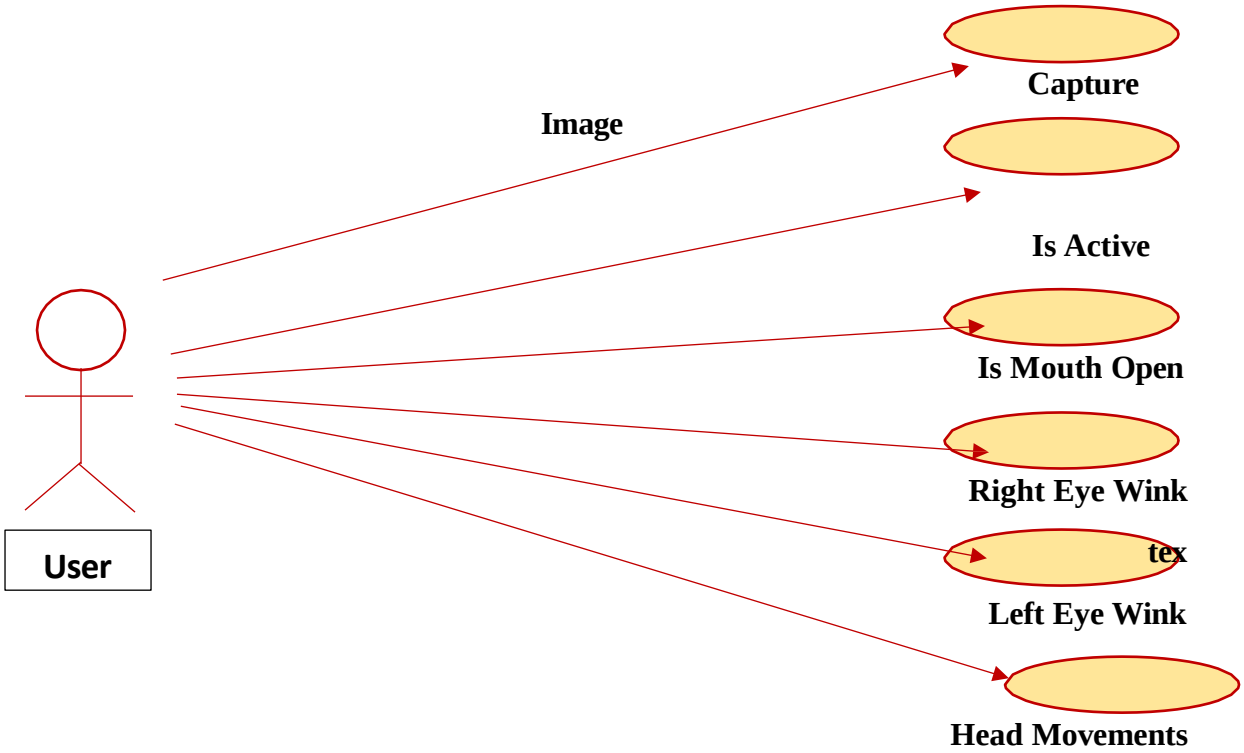


Figure 5.1 Use Case Diagram

5.3.2 Activity Diagram

Activity diagrams are the graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the UML, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system.

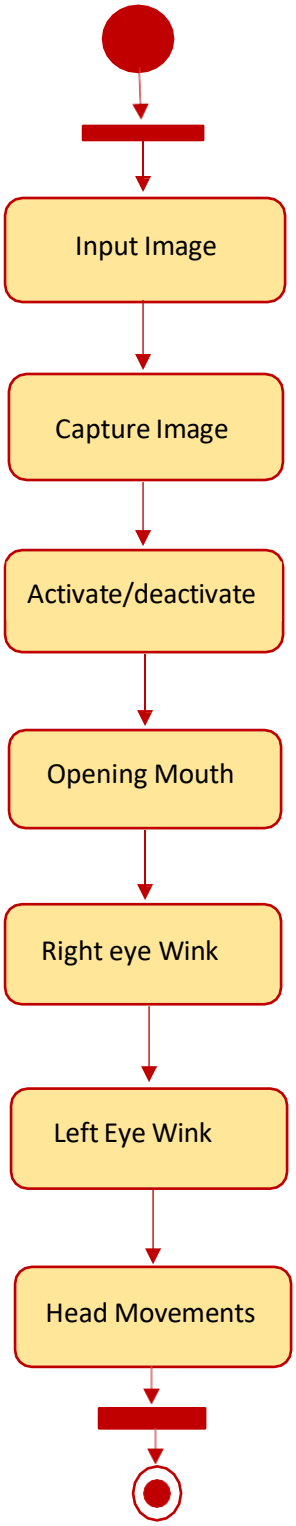


Figure 5.2 Activity Diagram

5.3.3 Sequence Diagram

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how to process operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario.

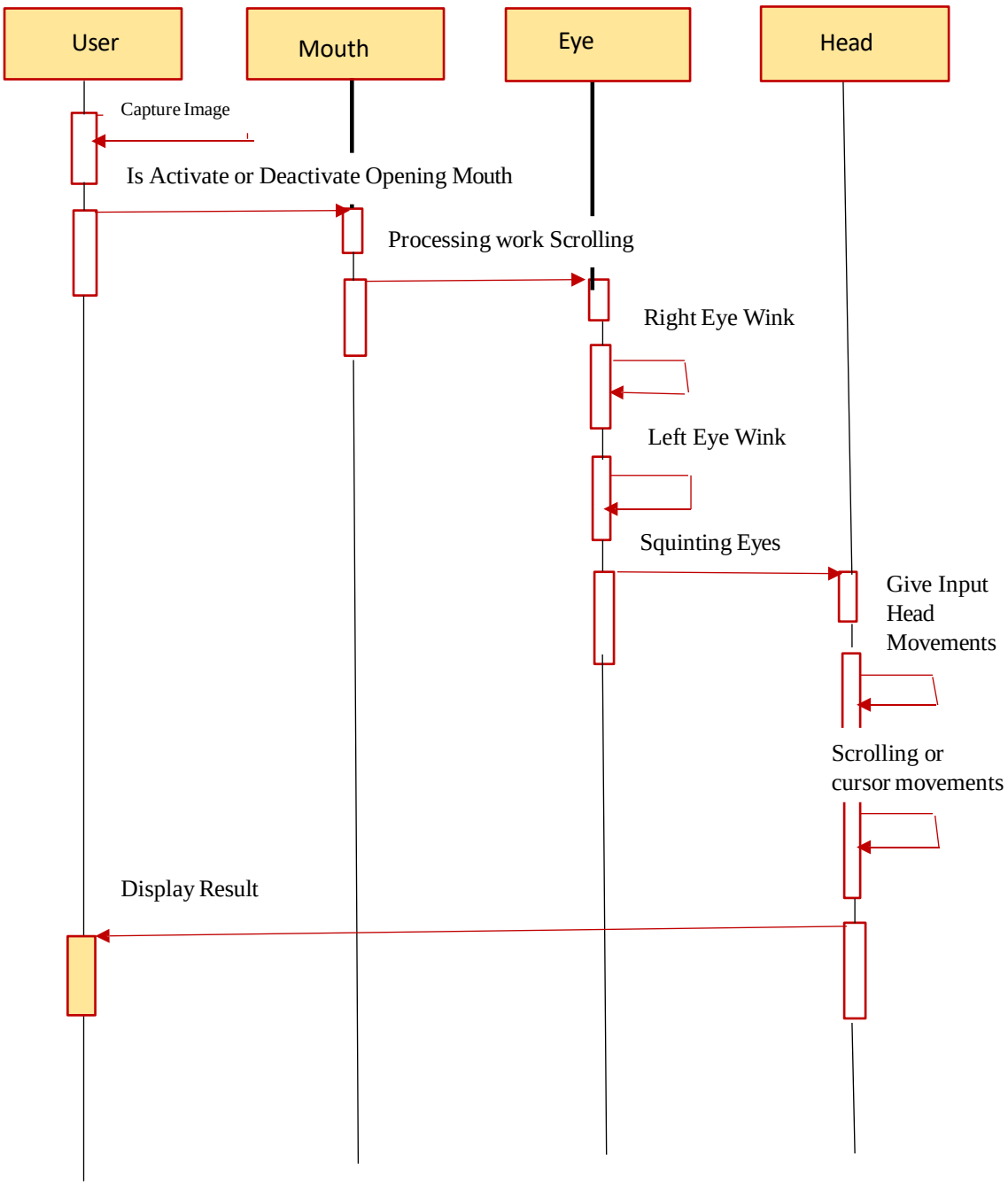


Figure 5.3 Sequence Diagram

5.3.4 Class Diagram

In this class diagram represents how the classes with attributes and methods are linked together to perform the verification.

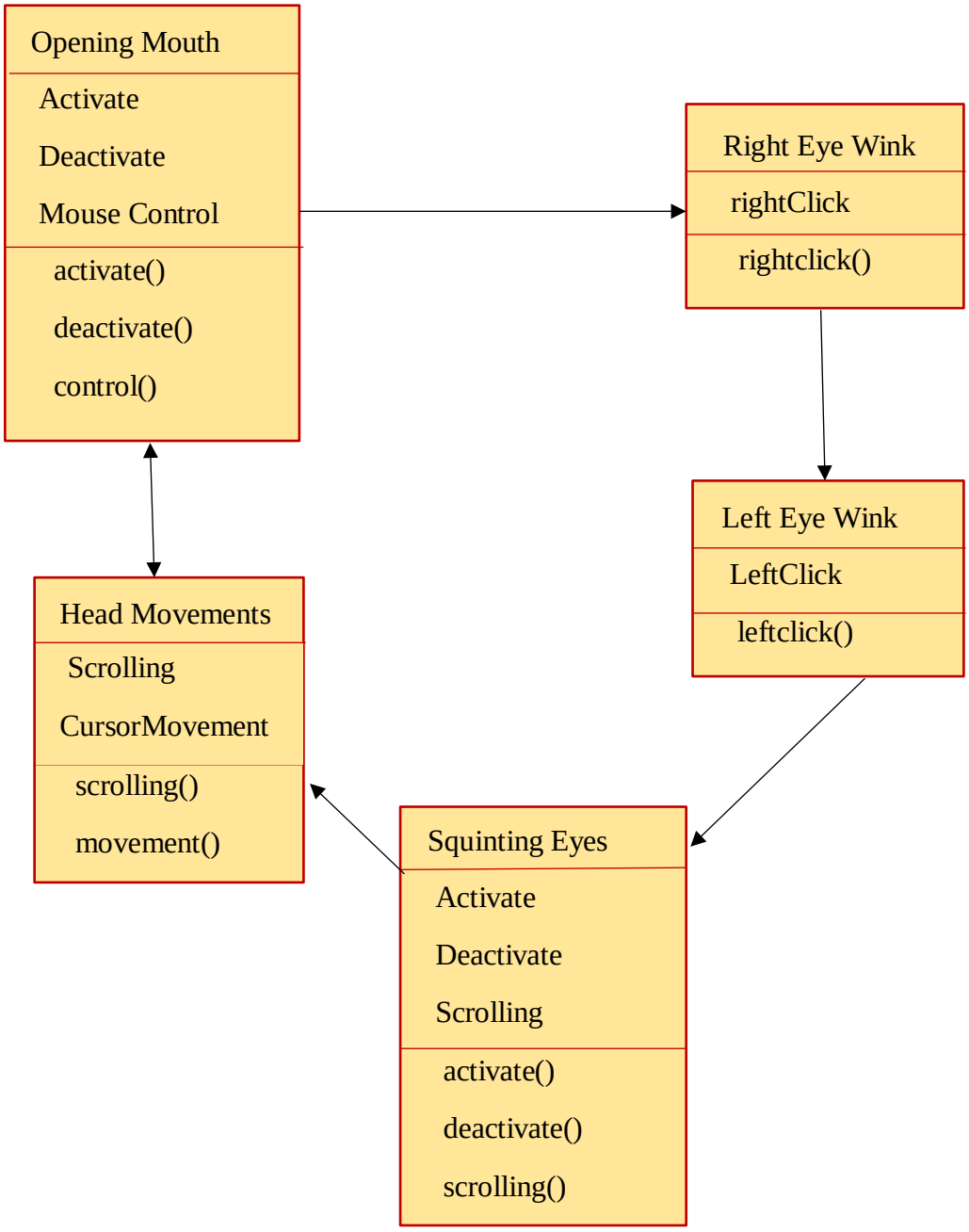
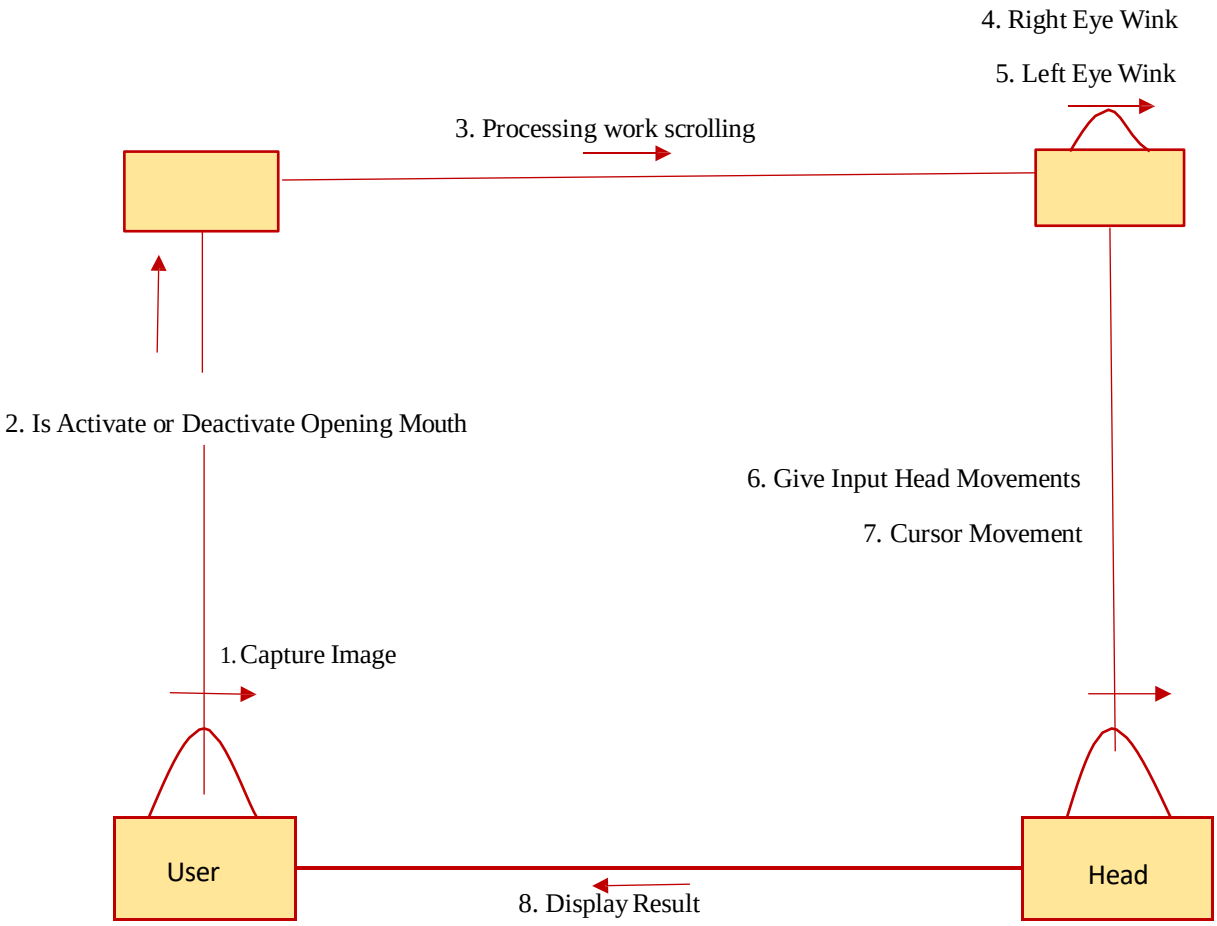


Figure 5.4 Class Diagram

5.3.5 Collaboration Diagram

A collaboration diagram, also called a communication diagram or interaction diagram, is an illustration of the relationships and interactions among software objects in the Unified Modeling Language (UML). The concept is more than a decade old although it has been refined as modeling paradigms have evolved.



5.5 Collaboration Diagram

5.3.6 Data Flow Diagram

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. Often, they are a preliminary step used to create an overview of the system which can later be elaborated. DFD's can also be used for the visualization of data processing (structured design).

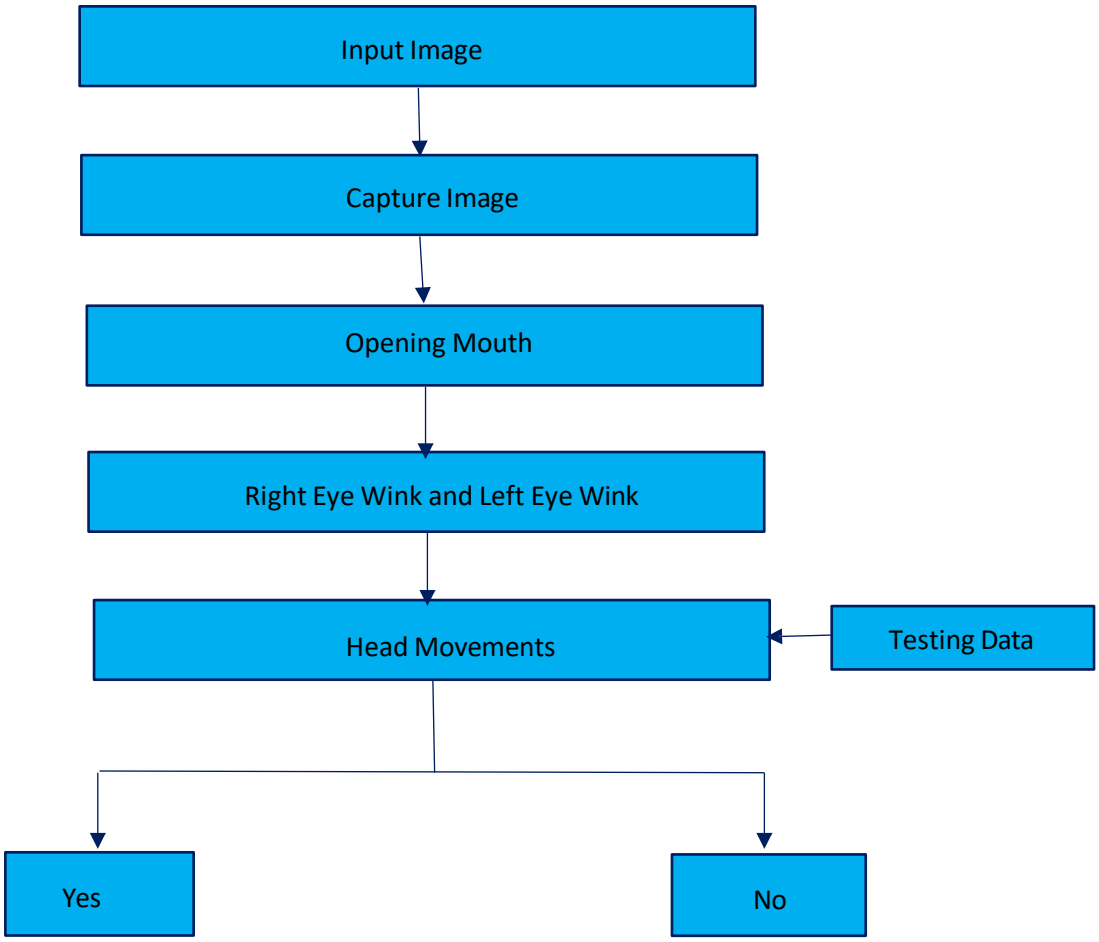
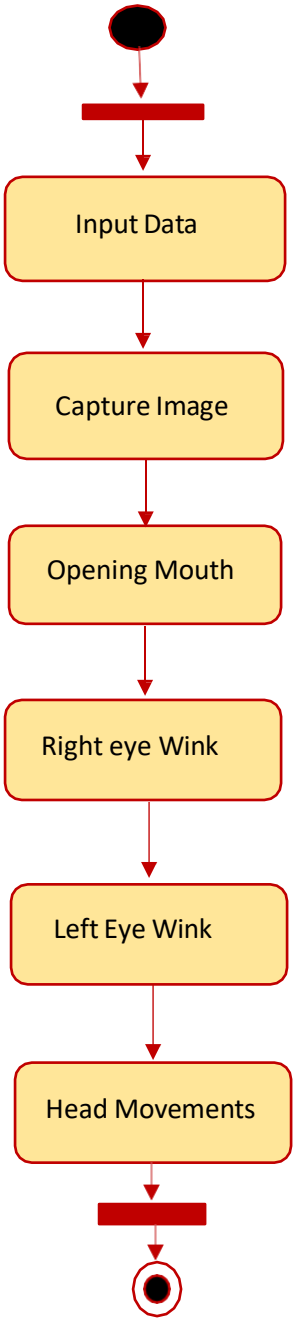


Figure 5.6 Data Flow Diagram

5.3.7 State Chart Diagram

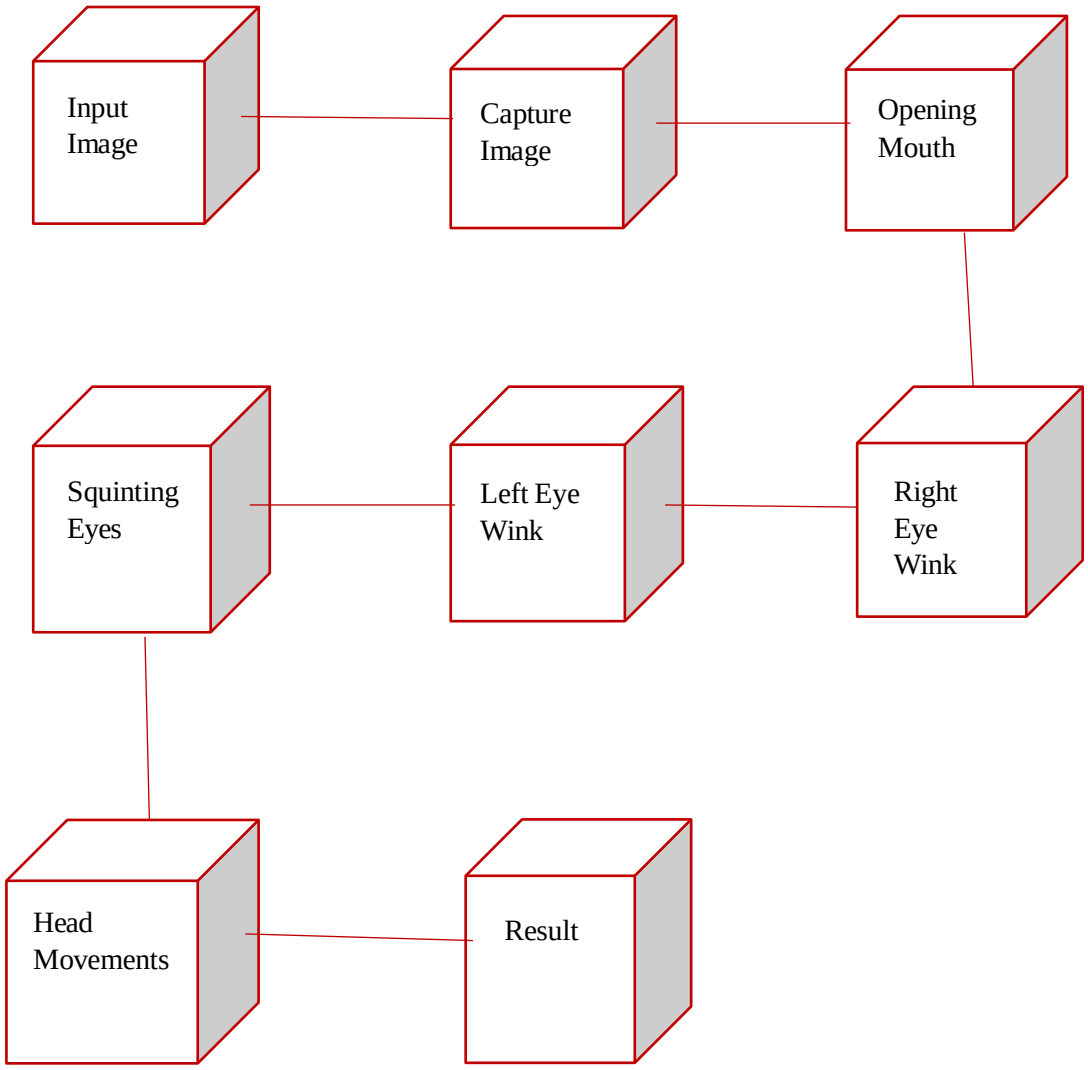
State diagrams are a loosely defined diagram to show workflows of stepwise activities and actions, with support for choice, iteration and concurrency. State diagrams require that the system described is composed of finite number of states; sometimes, this is indeed the case, while at other times this is a reasonable abstraction. Many forms of state diagrams exist which differ slightly and have different semantics.



5.7 State Chart Diagram

5.3.8 Deployment Diagram

In the Unified Modeling Language, a component diagram depicts how components are wired together to form larger deployment and or software systems. They are used to illustrate the structure of arbitrarily complex systems. User gives main query and it converted into sub queries and sends through data dissemination. Results are to be showed to user by data aggregators. All boxes are arrow indicates dependencies.



5.8 Deployment Diagram

CHAPTER 6

TESTING

6.1 GENERAL

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product it is the process or exercising components and user expectations and does not fail in an unacceptable manner. There its requirement of test. Each test type addresses a specific testing requirement. are various types of tests.

6.2 TEST PLAN

Any project can be divided into units that can be further performed for detailed processing. Then a testing strategy Tor can or this unit 1s carried out. Unit testing helps to identity the possible bugs in the individual component, so the component that has bugs can be identified and can be rectified from errors.

6.3 TEST CASES

The test case 1s initiated by developing comprehensive cases to test the general functionality and special Features on a variety of platform combinations. Strict quality control procedures are used. he processes verifies that the application meets the requirements specified in the system requirements document and is bug free. The following are the considerations used to develop the framework from test cases.

6.3.1 TYPES OF TESTS

1) Unit Testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program input produces valid outputs. All decision branches and

internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform configuration. Unit tests ensure that each unique path of a business process basic tests at component level and test a specific business process, application, and/or performs accurately to the documented specifications and contains clearly defined inputs and expected results.

2) Functional Test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input	:	Identified classes of valid input must be accepted.
Invalid Input	:	Identified classes of invalid input must be rejected.
Functions	:	Identified functions must be exercised.
Output	:	Identified classes of application outputs must be exercised.
Systems/Procedures	:	Interacting systems or procedures must be invoked.

3) System test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and knows, emphasizing pre-driven process links and integration points.

4) Performance Test

The Performance test ensures that the output is produced within the time limits, and the time taken by the system for compiling, giving response to the users and request being send to the system for to retrieve the results.

5) Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects. The

task of the integration test is to check that components in a software system or - one steps up-software that components or software applications, e.g. level -interact without error. applications at the company.

6) Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Acceptance testing for Data Synchronization

- The Acknowledgements will be received by the Sender Node after the Packets are received by the Destination Node
- The Route add operation is done only when there is a Route request in need
- The Status of Nodes information is done automatically in the Cache Updating process.

CHAPTER 7
IMPLEMENTATION

7.1 GENERAL

This chapter provides detail about the implementation of the system. It gives detailed view about the steps in implementation of the project.

7.1 SAMPLE CODE

```
import cv2
import mediapipe as mp
import pyautogui
cam = cv2.VideoCapture(0)
face_mesh = mp.solutions.face_mesh.FaceMesh(refine_landmarks=True)
screen_w, screen_h = pyautogui.size()
while True:
    _, frame = cam.read()
    frame = cv2.flip(frame, 1)
    rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    output = face_mesh.process(rgb_frame)
    landmark_points = output.multi_face_landmarks
    frame_h, frame_w, _ = frame.shape
    if landmark_points:
        landmarks = landmark_points[0].landmark
        for id, landmark in enumerate(landmarks[474:478]):
            x = int(landmark.x * frame_w)
            y = int(landmark.y * frame_h)
            cv2.circle(frame, (x, y), 3, (0, 255, 0))
            if id == 1:
                screen_x = screen_w * landmark.x
                screen_y = screen_h * landmark.y
```

```
            pyautogui.moveTo(screen_x, screen_y)
    left = [landmarks[145], landmarks[159]]
    for landmark in left:
        x = int(landmark.x * frame_w)
        y = int(landmark.y * frame_h)
        cv2.circle(frame, (x, y), 3, (0, 255, 255))
    if (left[0].y - left[1].y) < 0.004:
        pyautogui.click()
        pyautogui.sleep(1)
cv2.imshow('Eye Controlled Mouse', frame)
cv2.waitKey(1)
```

7.2 SCREENSHOTS

This project is implemented on application PyCharm using Python.

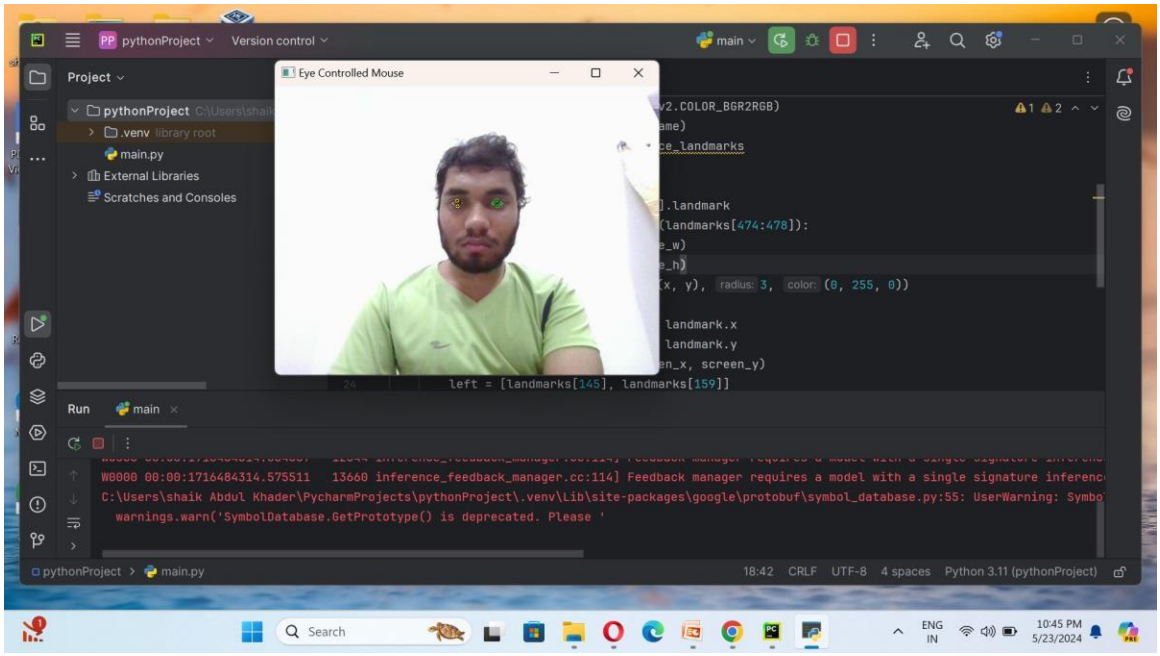


Figure 7.1 Eye Detection on Webcam

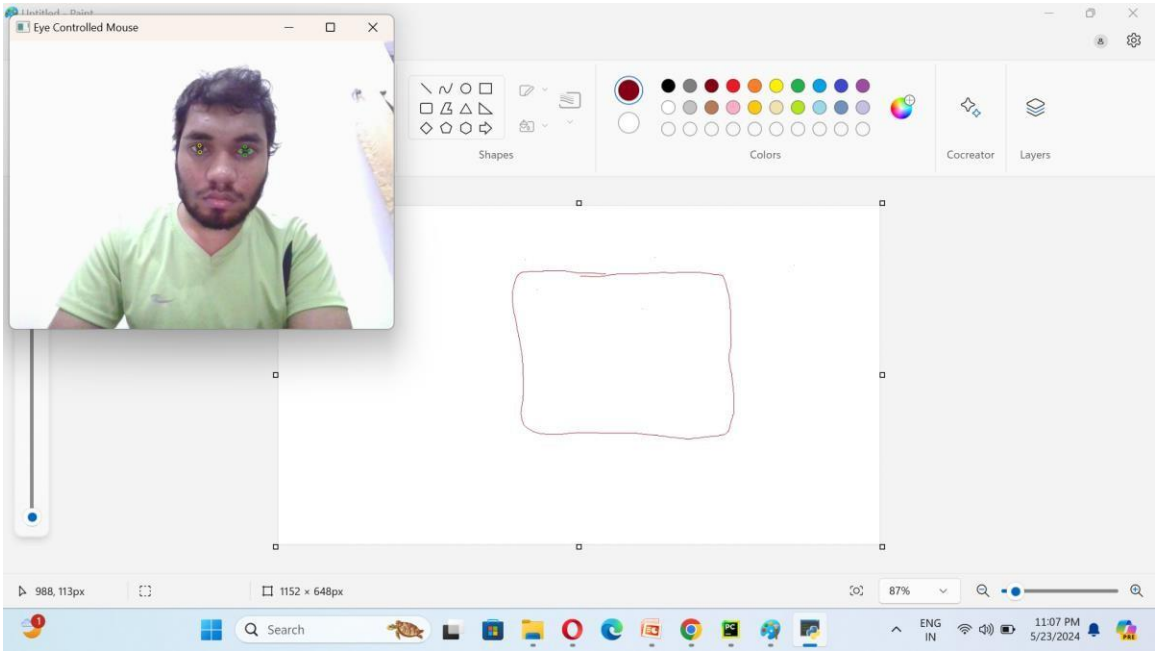


Figure 7.2 Drawing on Paint application using eye-ball based cursor movement

CHAPTER 8

APPLICATIONS AND FUTURE ENHANCEMENT

8.1 GENERAL

In this chapter, we are discussing about application and future enhancement of eye tracking technique. Here we are discussing few applications where we can use human and computer interaction application to find out a way to provide hands free interaction with the computer for the user.

8.2 APPLICATION

1. **Driver attention detection**

It is used to detect the drowsiness of the driver. It will alert the driver to take a break via an alert on.

2. **Gaming**

Eye tracking technology adds to the gameplay experience by allowing you to use your natural eye movements as an additional input in gaming, complementing keyboards, mice and gamepads.

8.3 FUTURE ENHANCEMENT

Further this system can be used in vehicles to observe the laziness of the driver by detecting the drowsiness symptoms like yawning. It can also be used in gaming and the Internet of Things domain. We can add the speech module which will be operated by user's mouse and launch on the start of the PC. Also, we can add scrolling functionality by using face movements and allow this application for playing games and banking sector also.

CHAPTER 9
CONCLUSION

A system that facilitates contactless control of mouse is developed and tested. It has an easy and wide application. Easy mouse cursor movement using face and switching between modes using eyes to enable or disable scrolling. Blinking to click. It has a wide range of applications. It can be used for physically disabled people in their education as this would empower them to type instead of writing without hand. This will lead to job opportunities for disabled people. The method described in this paper does not use any of this expensive equipment. Also, even physical connection has to be made with the user through any equipment. It just needs a computer with a webcam making it easy and cheap and uncomplicated method. By taking real-time images of the user the program is designed to evaluate these images and identify the type of action to be performed by taking a match against the previously stored expressions to identify the operation to be performed and take parameter if needed in case of moving the cursor the change in the relative position of eyes concerning the central position talks about the direction to move the cursor.

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